

Linear regression models to estimate NVIDIA's stock close price

To what extent can linear regression models reliably estimate NVIDIA's stock close price?

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1 Introduction:

While NVIDIA's extraordinary growth during the AI boom has captivated investors, accurately estimating the price of such volatile tech stocks remains one of the most challenging problems in quantitative finance. To do this, estimator variables (also known as parameters) are used. These may include related stocks, like TSMC or AAPL, to the stock price of small modular reactor companies like NuScale. This investigation therefore asks: to what extent can linear regression models reliably estimate NVIDIA's stock close price?

Although more advanced machine learning models, such as neural networks, can achieve high estimative accuracy, they require extensive computational resources, hyperparameter tuning, and mathematical depth that would shift the focus of this essay away from financial analysis. Therefore, linear regression models were selected due to their interpretability, methodological transparency, and suitability for evaluating the relative significance of estimator variables within the scope of this Extended Essay.

To address the question, seven linear regression models are constructed. These models are grouped into three categories: sector-linked variables, commodity-linked variables, and broader market indicators. Sector-linked variables includes major semiconductor firms and emerging small modular reactor companies. Commodity-linked variables include yearly metal production, Metal futures prices, and metal exchange-traded funds. Broader market indicators such as the Nasdaq 100 and the S&P 500. Each model uses lagged estimator variables to estimate NVIDIA's stock close price the following day. The resolution of metal production is yearly due to limitations in data availability.

Each model is evaluated using diagnostic and performance tests. The Breusch–Pagan and White tests are applied to detect heteroskedasticity, the Variance Inflation Factor (VIF) for multicollinearity and the Durbin-Watson test for Autocorrelation. Model performance is primarily compared using the coefficient of determination adjusted R^2 to measure explanatory power.

2 Methodology:

2.1 Linear Regression Models:

To estimate the stock price, Linear Regression Models (LRMs) are used. As Gujarati (2011) explains, LRMs are statistical tools used to model the interaction between data points. For example, one can use an LRM to model the relationship between the price of a good and the cost of raw materials. Or, as in the case of this investigation, one can use LRMs to model the relationship between NVIDIA's stock closing price and the numerical value of parameters that affect the stock. They work like this:

$$Y_i = \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \cdots + \beta_{ki} X_{ki} + u_i$$

Where Y is the independent variable—typically called the regressand—and can be measured in four different scales: ratio scale, interval scale, ordinal scale, and nominal scale. Ratio scale variables have 2 properties, (1) ratio of two variables, (2) distance between two variables, (3) ordering of variables. So, if Y takes 2 values, Y_1 and Y_2 , the ratio would be Y_1/Y_2 , distance would be $(Y_2 - Y_1)$, and the ordering or comparisons would be $Y_2 \leq Y_1$ or $Y_2 \geq Y_1$. Most variables belong to this category. Interval Scale are variables that don't satisfy the ratio property, an example of them would be distance between two years, (2025 - 2020); in this case (2025/2020) would not be meaningful. Ordinal Scale variables are those that don't satisfy the ratio property nor the distance property. For example, grading systems like A, B, C, D... Nominal scale variables are those that don't satisfy any of the properties, for example binary values like 1 or 0; these are often called dummy or categorical variables. As for X variables—typically called the regressors—they can also be measured in the same scales as the Y variable, though they are commonly measured in the ratio or interval scale. And lastly, the error term u is a stochastic error term that includes all the variables that could not be considered in the regressors. Meaning variables that can't be quantified or are not viable to quantify. (β is explained in the next section 2.2)

2.2 Ordinary Least Squares:

Ordinary Least Squares (OLS) is the method used to estimate the coefficients β in a linear regression model. It works by finding the values of β that minimize the sum of squared residuals — the differences between the observed values Y and the estimated values $\hat{Y}$. Mathematically, this is expressed as minimizing:

$$\sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Errors are squared rather than taken in absolute value to penalize larger errors more heavily and to ensure mathematical tractability. The solution to this minimization problem yields the OLS estimator:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

where X is the matrix of regressors and y is the vector of observed values. Under the Gauss-Markov assumptions — linearity, independence of errors, homoskedasticity, and no perfect multicollinearity — OLS produces the Best Linear Unbiased Estimator (BLUE), meaning no other linear unbiased estimator has a lower variance. Each of these assumptions is directly tested in this investigation through the Durbin-Watson, Breusch-Pagan, White's, and VIF tests respectively. (see next section for more information 2.3)

2.3 LRM Testing:

To be sure that the estimations made by the LRM are statistically significant, they have to be tested. This can be done through many different methods, but the main ones are the following: multicollinearity, heteroscedasticity, testing for the probability of the relation between the regressand and the regressor being a coincidence, and adjusted R^2 . All these values are interpreted against a significance threshold of 0.05 — meaning there is at most a 5% probability that the result occurred by chance, corresponding to a 95% confidence level. The probability of it being a coincidence is called the p-value; p-value is also used as a suffix to other values to convert them to probabilities. This P-value (in the case of testing for coincidence) is calculated by testing whether the slope could be

zero—meaning no significant relationship—using a t-test: assuming $y = \beta_0 + \beta_1 x + \epsilon$, then

$$t = \frac{\beta_1}{SE(\beta_1)}$$

(β_1 being the estimated slope and $SE(\beta_1)$ being the standard error of the slope). adjusted R^2 is the statistical measure of the proportion of variance in the regressand that can be explained by the regressor, meaning how well the regressor can explain the changes in the regressand. The heteroskedasticity or homoscedasticity tests for the variance of a variable's error.

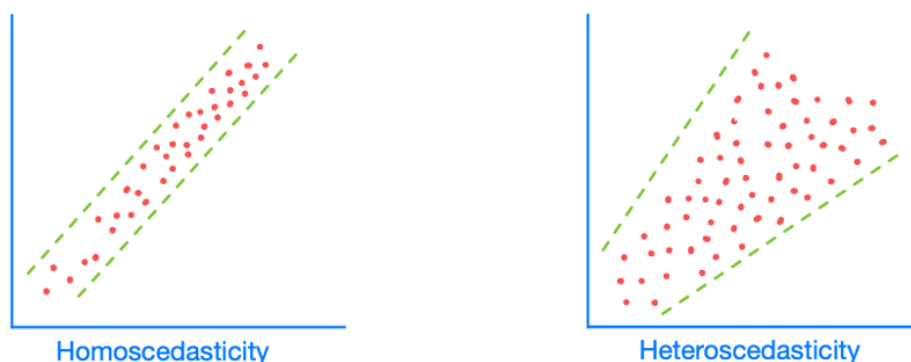


Figure 1: Illustration of homoscedasticity vs heteroscedasticity in regression residuals

Essentially, tests like White's test (a test for heteroskedasticity) test for variability of a variable being unequal across the range of values of a second variable that estimates it. This test returns a test-statistic that quantifies the degree of heteroskedasticity present in the model's residuals, which can then be converted to a p-value for interpretation. This test-statistic can also be turned into a P-value—called a test-statistic p-value that is used to interpret the data easily. Furthermore, Multicollinearity tests how related the variables are. For example, if you have $X_1 = \#$ of hours taught and $X_2 =$ number of classes, both these variables are redundant. If the variables are too related, then the coefficients for each of them will be inaccurate and unreliable.

In figure 2, multicollinearity is seen quite well. There are 2 distinct groups of variables, X2-4 and X1 and X5. These groups have very similar values and thus are somewhat

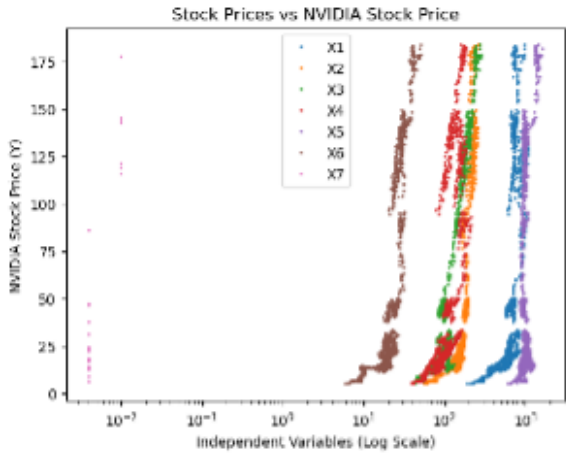


Figure 2: Example of high multicollinearity between regressors

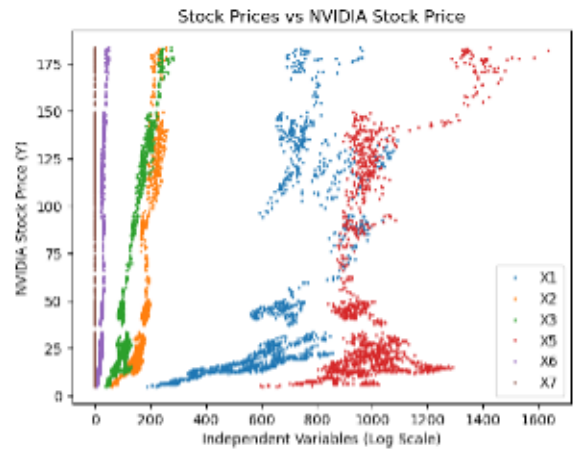


Figure 3: Example of low multicollinearity between regressors

redundant in the regression model. In figure 3, there is less multicollinearity, though it can still be better. Finally, the Durbin-Watson test checks whether the residuals of a regression model are correlated with each other across observations (serial autocorrelation). It produces a statistic ranging from 0 to 4, where values near 2 indicate no autocorrelation, values near 0 indicate positive autocorrelation, and values near 4 indicate negative autocorrelation. Finally, the Cochrane-Orcutt procedure is an iterative method used to correct for autocorrelation in OLS regression. It works by first estimating the autocorrelation coefficient ρ from the residuals of the original model, then transforming all variables by subtracting ρ times their lagged value:

$$Y_t^* = Y_t - \rho Y_{t-1} \quad \text{and} \quad X_t^* = X_t - \rho X_{t-1}$$

OLS is then re-applied to the transformed variables, producing corrected coefficient estimates with more reliable standard errors. A successfully corrected model should return a Durbin-Watson statistic close to 2. Overall, to ensure reliable coefficients, Heteroskedasticity, Multicollinearity, and the P-value for coincidence needs to be less than 0.05, the adjusted R^2 must be more than 0.95.

3 Model Selection:

A wide range of potential explanatory variables could plausibly influence NVIDIA’s stock price, including input costs, sector performance, macroeconomic indicators, and energy-market dynamics. Each group captures a distinct theoretical transmission mechanism — supply-chain dependence, commodity cost exposure, energy demand pressures, or overall market sentiment.

The objective is not merely to maximize R^2 , but to evaluate which category of variables produces the most statistically reliable linear model when subjected to diagnostic testing. Accordingly, each parameter group is assessed using the heteroskedasticity, multicollinearity, autocorrelation, and goodness-of-fit tests outlined in Section 2.3.

To preserve replicability and methodological transparency — both essential in quantitative research — all datasets were selected according to three criteria: public availability, free access, and ease of independent retrieval. This ensures that the investigation can be reproduced by any researcher with access to a computer and internet connection, strengthening the academic validity of the study.

To ensure systematic comparison, the variables in this investigation are organized into seven structured parameter groups in the following table. An eighth group is then introduced as a synthesis of the best parameters.

Model 1:		Model 3:		Model 5:	Model 6:	Model 7:
Related		Daily Metal				Modular
Stocks	Model 2:	Futures	Model 4:	S&P 500	Nasdaq	Nuclear
(Ticker)	Yearly	Prices	Monthly			Reactors
	Metal		Electricity			
	Production		Rates			
	(Metric Tonnes)		(USD/kWh)			
AMD	Copper	Copper	US	S&P 500	Nasdaq	NuScale
			Industrial			
AAPL	Gold	Gold	US			Oklo
			Commercial			
ASML	Silver	Silver				GE Vernova
TSMC	Silicon	Platinum				BWX
GOOG	Tin					Nano
						Nuclear
QCOM	Aluminum					
INTC	Rare					
	Earths					

3.1 Model Analysis

3.1.1 Model 1:

Model 1, related stocks, is useful but not perfect. The coefficients are:

Stock	Coefficient
ASML	-0.0432
AAPL	0.4330
TSMC	0.6600
QCOM	-0.0197

Table 1: Regression Coefficients for Semiconductor Stocks

The adjusted R^2 of 0.916 suggests a strong link between the regressors and the re-

gressand, further strengthened by a Pval of less than 0.001. The Breusch-Pagan (LM = 523.46, $p < 0.001$) and White's test ($\chi^2 = 909.06$, $p < 0.001$) reject the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. The Durbin-Watson statistic of 0.097 ($p \approx 0.0$) indicates strong positive autocorrelation in the residuals, violating the OLS independence assumption — likely due to the time-series nature of stock price data. Coefficient estimates remain unbiased, but standard errors and hypothesis tests may be unreliable. Additionally, the VIFs are quite high for TSMC:

Stock	VIF score
ASML	3.993619
AAPL	5.319020
TSMC	4.362418
QCOM	3.110856

Table 2: VIF scores for Semiconductor Stocks

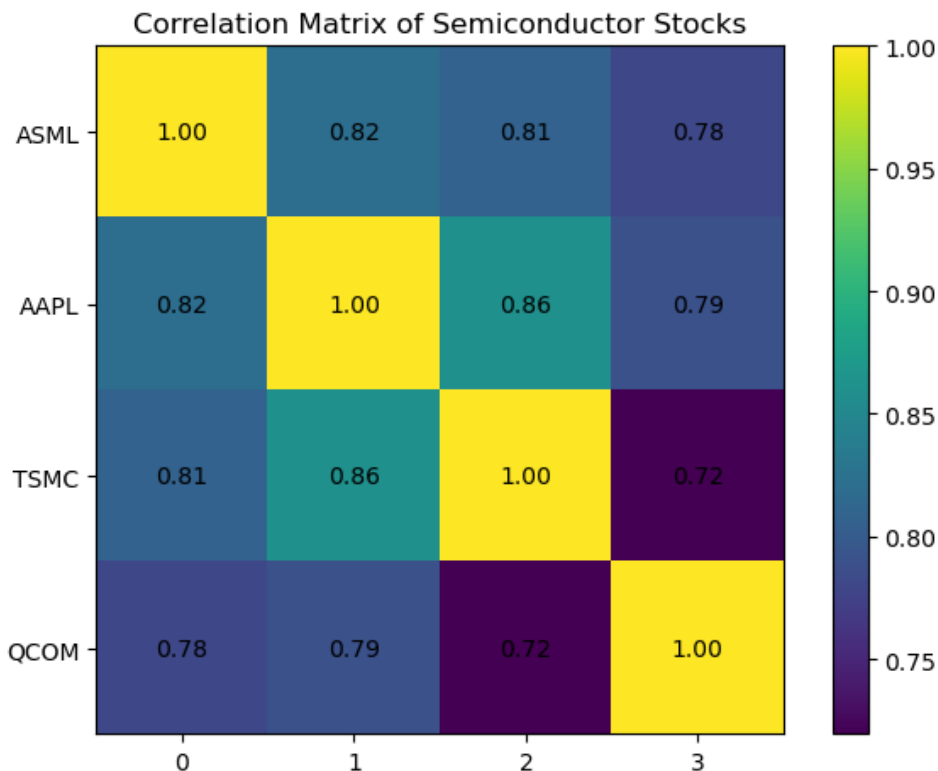


Figure 4: Correlation matrix of semiconductor stocks used in Model 1

Meaning that one or more variables are co-linear and thus redundant. Overall, using related stocks to estimate NVIDIA's stock price is a decent way to estimate in which

direction NVIDIA's stock price might go, but not definitive enough to estimate an actual future price. Additionally, it would be necessary to use more stocks and conduct a proper analysis on related stocks to ensure the VIFs stay low. In Fig 4, the stock with the highest correlation is and TSMC at 0.86. Although the other correlations are still quite high, ranging from 0.72 to 0.82. A possible reason for high VIFs is the fact that Intel relies on TSMC for chip manufacturing. Furthermore, high overall VIFs could be caused by an interdependence of most tech companies on each other.

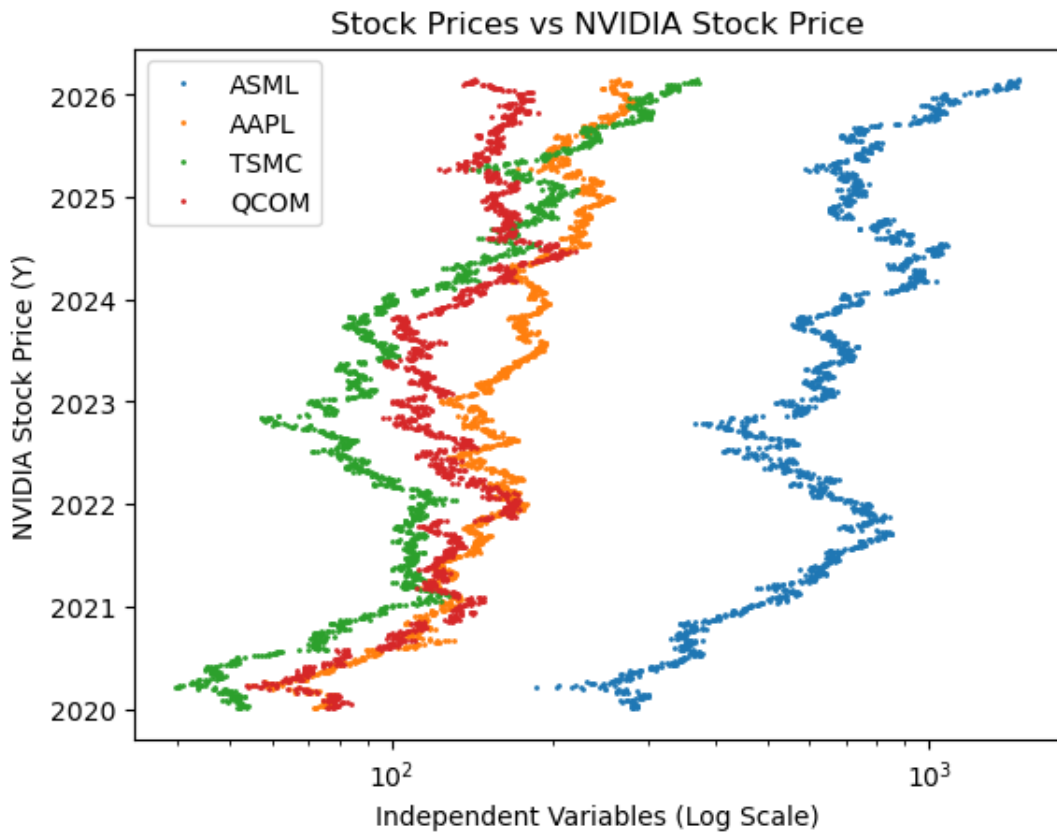


Figure 5: VIF plot for Model 1 regressors

3.1.2 Model 2:

Model 2, yearly metal production, includes metals the USGS reports on and that TSMC uses to produce NVIDIA GPUs. They have very low coefficients and thus is likely not great to estimate NVIDIA's future stock price, at least not on its own.

Metal	Coefficient
Silver	-0.0001
Tin	-9.472e-07
Silicon	-7.556e-07
Aluminum	1.33e-07
Rare Earths	3.648e-05
Gold	0.0020

Table 3: Regression Coefficients for Semiconductor Stocks

The adjusted R^2 is 0.815, lower than the ideal 0.95 (for an error of 0.05 or 5%) meaning a linear model is not the best fit for this data. Similarly, the Breusch-Pagan and White's tests results are good. The result of the Breusch-Pagan (LM = 11.14, p = 0.084) and White's test ($\chi^2 = 19.0$, p = 0.39) accept the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. The Durbin-Watson statistic of 2.586 (p = 0.2904) provides no strong evidence of autocorrelation, suggesting the independence assumption holds reasonably well for this model. However, the VIFs are quite high.

Metal	VIF
Silver	17.446360
Tin	2.037078
Silicon	28.393788
Aluminum	66.044190
Rare Earths	2.597244
Gold	10.959539

Table 4: VIFs for yearly metal production

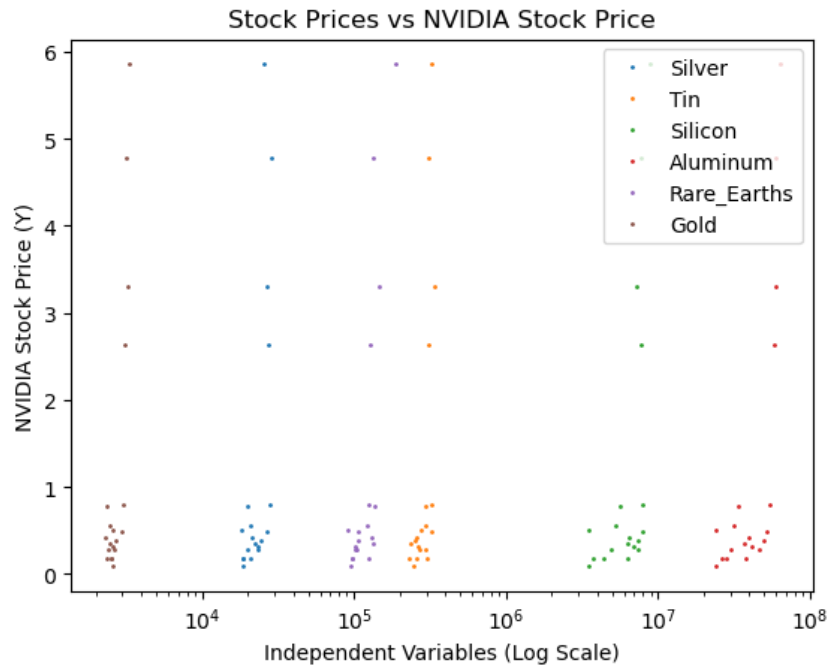


Figure 6: Scatter plot of yearly metal production vs NVIDIA close price

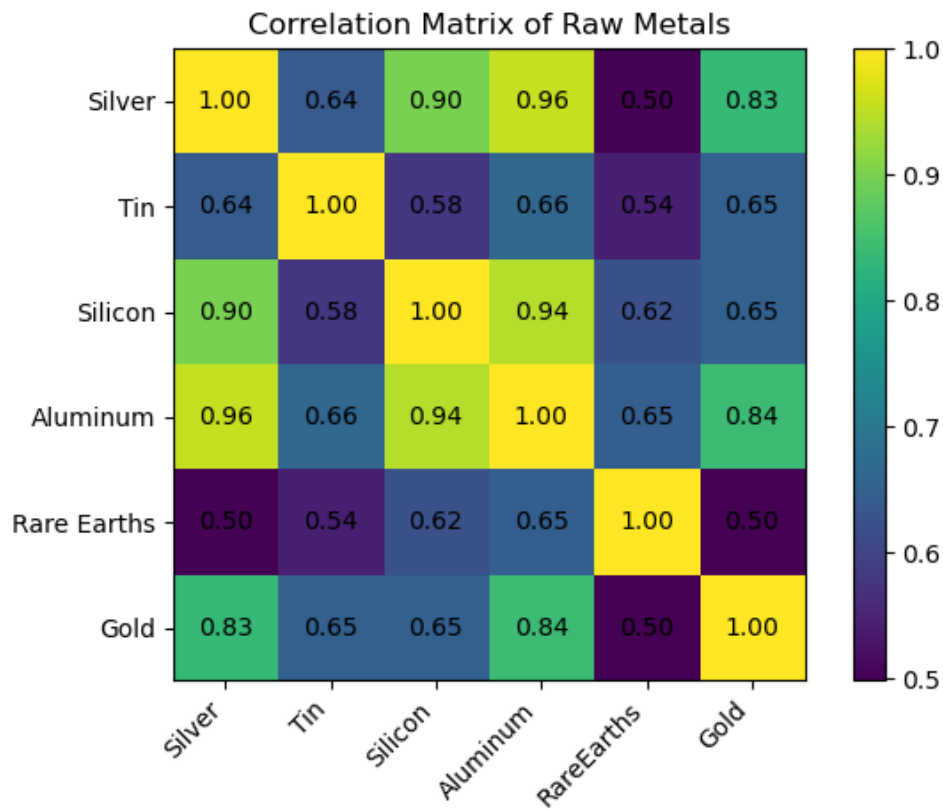


Figure 7: Correlation matrix of yearly metal production variables used in Model 2

These high VIFs could be caused by high correlation coefficients. In Fig 7 it is possible

to see that the correlation coefficient between aluminum and silicon, and aluminum and silver, are quite high at 0.94 and 0.96 respectively; followed by aluminum and gold at 0.84. This possibly explains why aluminum has the highest VIF, as seen in table 4, at 66.04 and why gold, silver, and silicon are also high at 10.96, 17.45, and 28.39 respectively.

3.1.3 Model 3:

Model 3 consists of the daily metal futures prices for copper, gold, silver, uranium, and platinum. Chosen for their availability and relevance in GPU production or electricity production in the case of uranium. A futures contract is an agreement to buy something, in this case a metal, at a set price on a set date. For example, if a buyer agrees to buy 1 tonne of copper a week in the future for 13,000 USD, the buyer can either have a gain if the price goes over 13,000 USD at the agreed buy date or have a loss if the price goes under 13,000 USD at the agreed buy date. A bet on profit (gain) is called a speculation and when a company buys a futures contract to lock in the price it is called hedging. TSMC, for example, NVIDIA’s chip supplier, could buy a futures contract for copper, gold, silicone, etc to lock in prices in the future.

Metal Futures	Coefficient
Copper	12.8265
Gold	0.0941
Silver	-0.6901
Platinum	-0.0695

Table 5: Regression Coefficients for Metals Futures

In this case the model is comparing daily close price. So, for example, if copper changes by \$1 NVIDIA changes by \$12.8265. The model has an adjusted R^2 of 0.898, which is slightly lower than the ideal of 0.95. Furthermore, the Breusch-Pagan (LM = 512.00, $p < 0.001$) and White’s test ($\chi^2 = 683.22$, $p < 0.001$) reject the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. The Durbin–Watson statistic of 0.034 signals severe serial correlation in the residuals ($p \approx 0.0$), violating OLS independence assumptions. Standard errors and significance tests are therefore unreliable, though coefficient estimates remain unbiased.

Metal Futures	VIF
Copper	2.109996
Gold	7.038074
Silver	18.195851
Platinum	8.528261

Table 6: Regression Coefficients for Metals Futures

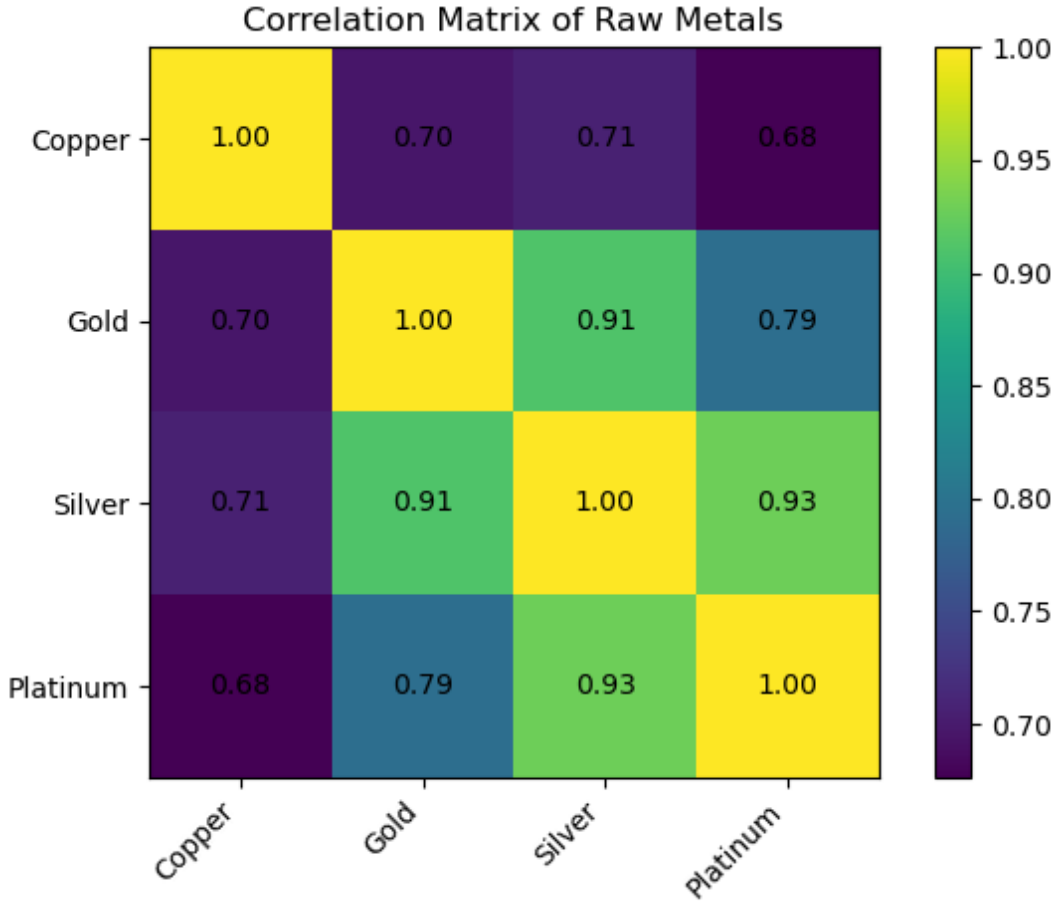


Figure 8: Correlation matrix of daily metal futures prices used in Model 3

The high VIFs, particularly silver's, could be explained by several factors. First, silver, gold, and platinum are all part of the precious metals industry and thus tend to move together. Secondly, in Fig 8, silver has the highest correlation coefficients at 0.93 with platinum and 0.91 with gold. Then, copper's low VIF, could be explained by first, it has the lowest correlation coefficients in Fig 8, but also, it is not part of the precious metals market meaning that it appeals to other buyers and thus doesn't move with it, making it unique in this group.

3.1.4 Model 4:

Model 4 consists of metal ETFs. ETF means Exchange-Traded Fund. It is a type of investment fund that holds a collection of assets. GLD, SLV, and PPLT hold gold, silver, and platinum ingots, respectively; CPER holds copper futures contracts; and URA holds stocks of uranium mining companies.

Metal ETFs	Coefficient
Gold	0.7520
Silver	-0.3677
Copper	0.4037
Platinum	-0.8809
Uranium	2.4528

Table 7: Regression Coefficients for Metal ETFs

These coefficients have an adjusted R^2 value of 0.927. This means that about 92.7% of NVIDIA's stock movements can be explained by these metal ETFs. The metal with the biggest effect on NVIDIA's stock is Uranium. Likely because it is quickly becoming crucial for IAs. AIs need very high amounts of energy to operate and the current electric grids are struggling to meet those demands; because of this, many companies are developing modular nuclear reactors to power these IA super computers; and some of those reactors need uranium to run. The other metals are used in the production of GPUs, NVIDIA's main product. Both the Breusch-Pagan (LM = 321.64, $p < 0.001$) and White's test ($\chi^2 = 555.78$, $p < 0.001$) reject the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. The Durbin-Watson statistic (0.049, $p \approx 0.0$) reveals strong autocorrelation in the residuals, which is unsurprising given the time-series nature of stock prices. This means hypothesis testing results should be interpreted with caution.

Metal ETFs	VIFS
Gold	9.303403
Silver	17.051093
Copper	3.061417
Platinum	7.710108
Uranium	6.072512

Table 8: Regression Coefficients for Metal ETFs

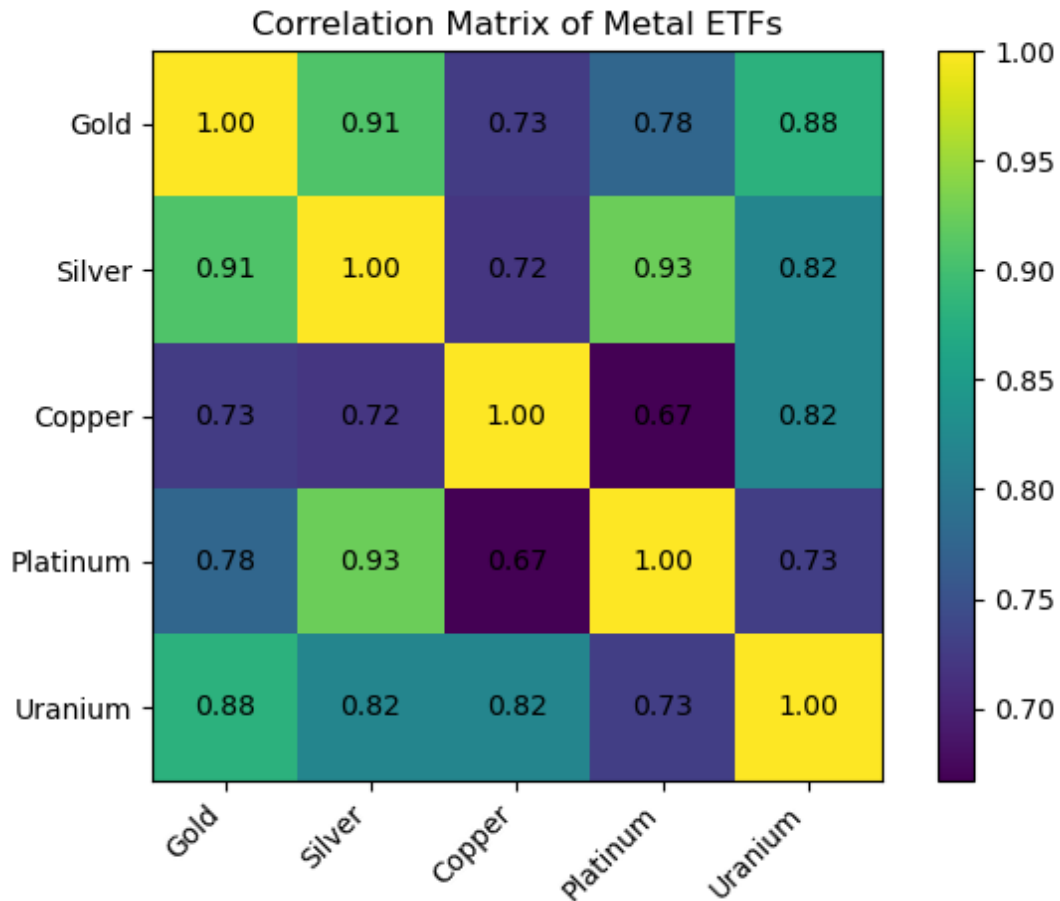


Figure 9: Correlation matrix of metal ETFs used in Model 4

The high VIFs, seen in Table 8, for gold, silver, and platinum, as in group 3, can be attributed to them being in the same industry and them having the highest correlation coefficients between them. Then, uranium's high VIF at 6, could possibly be explained by its correlation coefficients mostly staying in the 0.8-0.9 range except for platinum which is lower at 0.73. And lastly, as seen in group 3 as well, copper is in a separate industry to all the other metals plus it has the lowest overall correlation coefficients.

3.1.5 Model 5:

The S&P 500 is a stock index that tracks the 500 largest US companies, weighted by market cap. It is widely considered the best single measure of large-cap US companies and an overall measure of market health. Compared to other indexes, it also serves as the best measure of overall US economic health. This can be compared to the NVIDIA

stock although currently NVIDIA is part of the 500 companies. For this model, this investigation uses the daily percentage change of both stocks

Index	Coefficient
S&P 500	0.0524

Table 9: Regression Coefficient for S&P 500

The coefficient of 0.0524 means that for every \$1 the S&P 500 moves, NVIDIA will move 0.0524\$ in the same direction (+ or -). The adjusted R^2 of 0.901 means that about 90% of NVIDIA's stock movements can be explained by the S&P 500's movements. Which for a single stock is incredibly high. A Breusch-Pagan test revealed significant heteroskedasticity in the model (LM = 142.03, $p < 0.001$) and a White's test revealed even stronger evidence ($\chi^2 = 369.59$, $p < 0.001$), this means the model rejects the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. A Durbin-Watson statistic of 0.040 ($p \approx 0.0$) reveals strong autocorrelation in the residuals. However, the VIF test returned a value of 1, although this is expected since only one stock is being compared to NVIDIA. Despite a high R^2 of 0.901, the severe autocorrelation (DW = 0.040) renders Model 5 unreliable for estimating NVIDIA's close price, as its explanatory power is trend-driven rather than genuinely predictive.

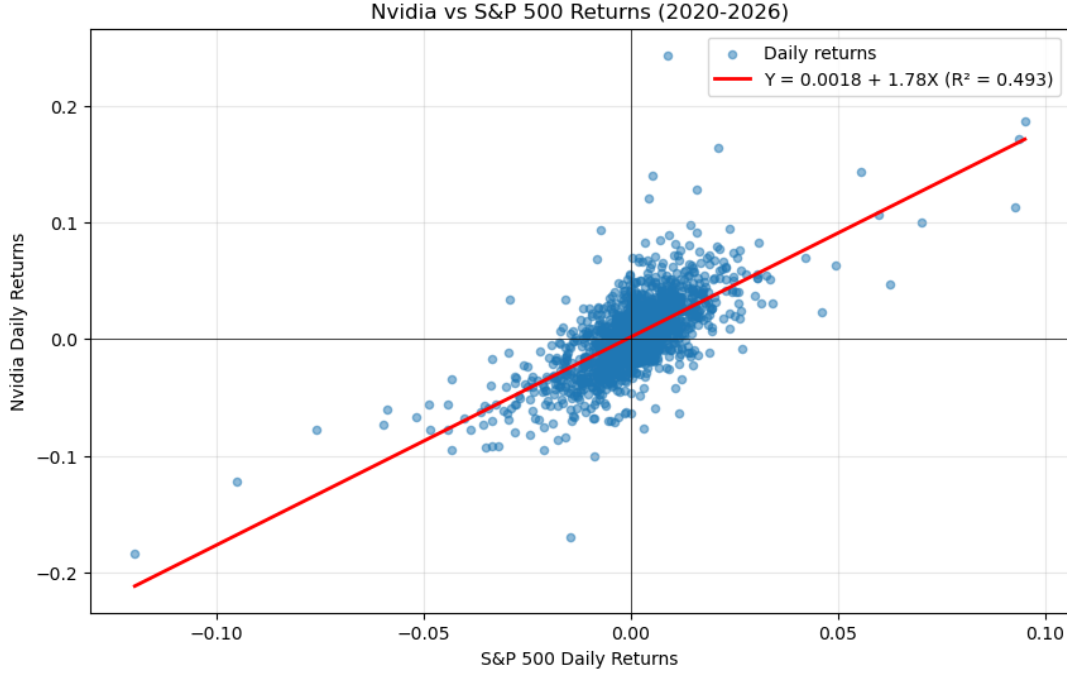


Figure 10: Scatter plot of S&P 500 vs NVIDIA daily close price

3.1.6 Model 6:

The NASDAQ 100 index represents all stocks and companies traded on the NASDAQ exchange. It works similar to the S&P 500, each company has a weight assigned to it based on its market value. For example, NVIDIA's weight is currently (3/2/2026) 12.96%. But in difference to the S&P 500, the NASDAQ mostly centers around tech companies, making it more suitable for estimating NVIDIA's daily close price since NVIDIA is a major component of the index and shares similar market drivers with other tech stocks in the index. Furthermore, the adjusted R^2 returned a value of 0.910, meaning that 91% of NVIDIA's changes can be explained by NDX.

Index	Coefficient
NDX	0.0124

Table 10: Regression Coefficient for NASDAQ 100

The coefficient of 0.0124 means that for every \$1 change in NDX, NVDA changes by 0.0124\$ in the same direction (+ or -). Furthermore, the Breusch-Pagan (LM = 81.89, $p < 0.001$) and White's test ($\chi^2 = 131.77$, $p < 0.001$) both strongly reject the

null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. The Durbin–Watson statistic of 0.045 ($p \approx 0.0$) provides strong evidence of autocorrelation, suggesting the independence assumption doesn’t hold well for this model. However, the VIF test returned a value of 1, which is expected since only one variable is being compared to NVIDIA. Despite a high R^2 of 0.910, the severe autocorrelation ($DW = 0.045$) renders Model 6 unreliable for estimating NVIDIA’s close price.

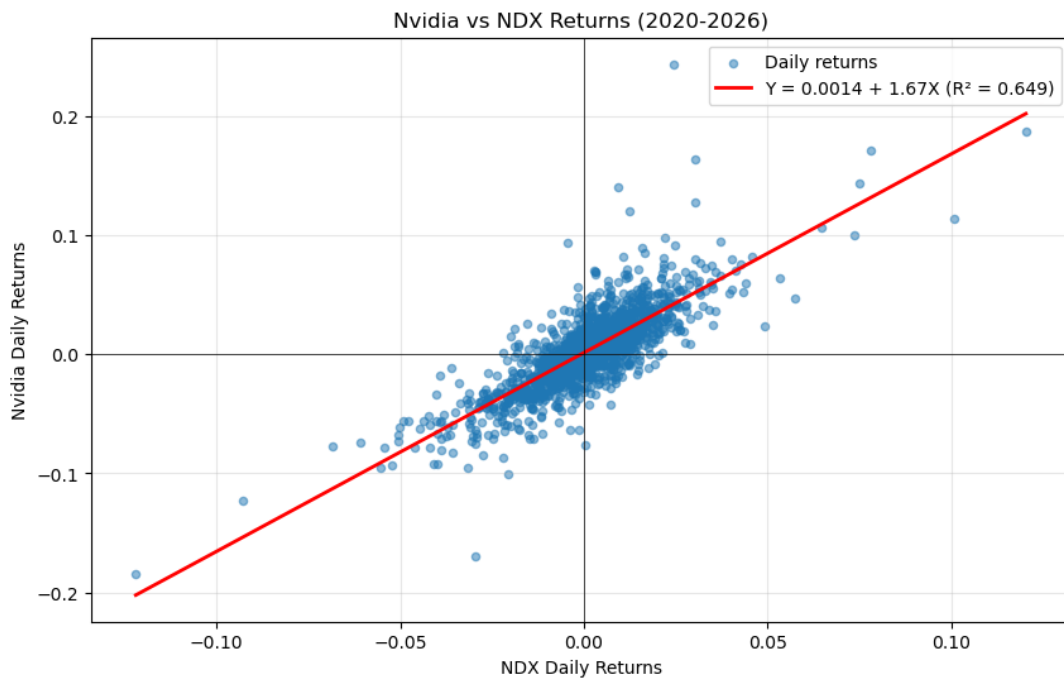


Figure 11: Scatter plot of NASDAQ 100 vs NVIDIA daily close price

3.1.7 Model 7:

AI data centers need high amounts of electricity, but current electric grids are struggling to meet the demand. Because of this, many AI companies, like Google, Meta, OpenIA, etc, are investing in modular nuclear reactors to power their data centers. There are many companies that make these, for example Oklo. Oklo is an American company founded in 2013 that focuses on developing small modular reactors (SMRs up to 300 MW). Another example is Nano nuclear, a company that develops micro modular reactors (MMR typically under 10-20 MW). These reactors would be placed next to or near data

centers to power them while at the same time also isolating them from the greater grid which can be vulnerable to outages which can cost AI companies billions of dollars.

SMR and MMR Company Stocks	Coefficients
NuScale	0.3682
Oklo	-0.0811
GE Vernova	0.0176
BWX	0.6739
Nano Nuclear	0.0078

Table 11: Coefficients for SMR and MMR Company Stocks

These coefficients, along with an adjusted R^2 of 0.906 shows a decent relationship between these stocks and NVIDIA's stock. Furthermore, both the Breusch-Pagan (LM ≈ 49.07 , $p < 0.001$) and White's test ($\chi^2 = 138.58$, $p < 0.001$) strongly reject the null hypothesis of homoskedasticity. Robust standard errors (HC3) were used to account for heteroskedasticity. A key limitation of this model is the Durbin-Watson statistic of 0.170 ($p \approx 0.0$), suggesting positive autocorrelation in the residuals. This is a known consequence of applying OLS to time-series data, and implies that reported standard errors may understate true uncertainty.

SMR and MMR Company Stocks	VIFS
NuScale	3.252240
Oklo	9.275726
GE Vernova	7.662576
BWX	10.590868
Nano Nuclear	5.090562

Table 12: VIFs for SMR and MMR Company Stocks

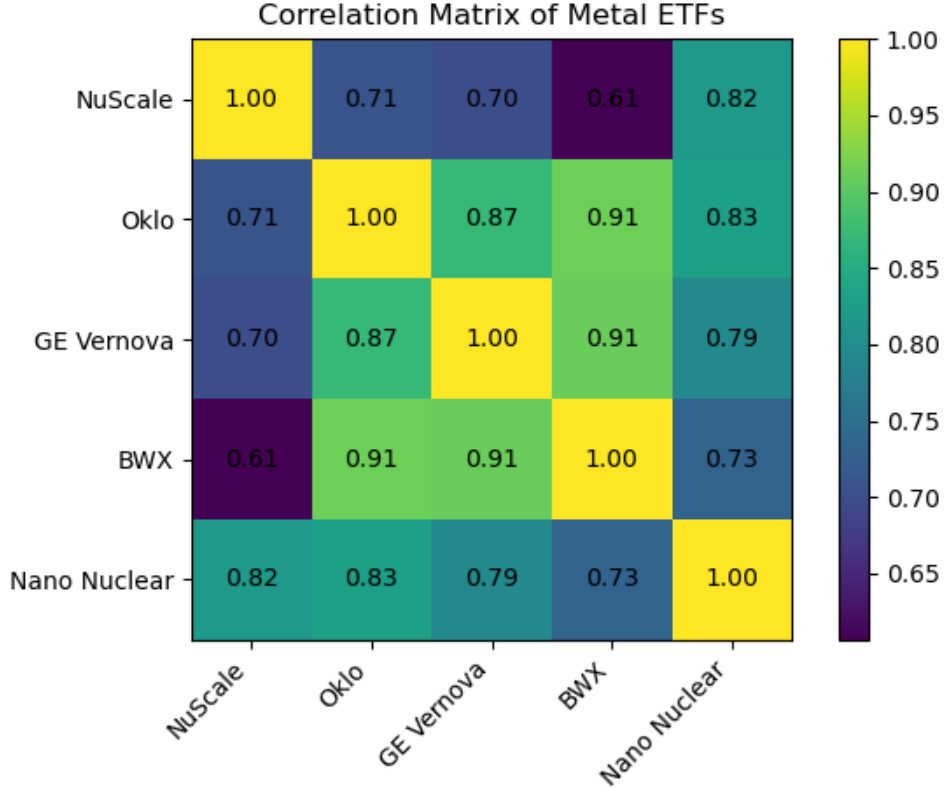


Figure 12: Correlation matrix of SMR and MMR company stocks used in Model 7

The high VIFs for these stocks stem from the fact that they are all in the same niche industry. Furthermore, they all respond to the same nuclear policy news, uranium prices, and energy sector trends. NuScale's low VIF could be explained by the fact that they are the most developed of the companies, being the only ones to have gotten Nuclear Regulatory Commission approval. Furthermore, BWX has the highest VIF and correlation coefficients; while NuScale has the lowest.

3.1.8 Model Composite: A Synthesis of Optimal Parameters

Model Composite consists of a mix of parameters from models 1, 4 and 6.

Parameter	Coefficient
ASML	-0.0803
Uranium ETF	1.2540
Nasdaq 100	0.0131

Table 13: Model Composite Coefficients with Newey-West Standard Errors

These 3 parameters were found to be the best due to their acceptable VIFs, high

adjusted R^2 (0.932), and perfect Breusch-Pagan and White's test results. The uranium ETF's coefficient (1.254) dominates, reflecting AI's energy demands, while NASDAQ captures tech sector sentiment and ASML shows negative correlation possibly from portfolio rebalancing. The Breusch-Pagan (LM = 187.45, $p < 0.001$) and White's test ($\chi^2 = 609.77$, $p < 0.001$) strongly reject the null hypothesis of homoskedasticity. Newey-West (HAC) standard errors with 8 lags were applied to account for both heteroskedasticity and autocorrelation. All coefficients remain highly significant ($p \leq 0.001$). The Durbin-Watson statistic of 0.024 indicates severe autocorrelation, common in financial time series. While Newey-West standard errors correct for this in our hypothesis tests, time series models would be better for out-of-sample forecasting.

Parameter	VIF
ASML	4.179405
Uranium ETF	7.562880
Nasdaq 100	7.107493

Table 14: VIFs for Model Composite

The VIFs are acceptable. Ideally, they would be lower (closer to 1), but for the scope of this essay they are sufficient.

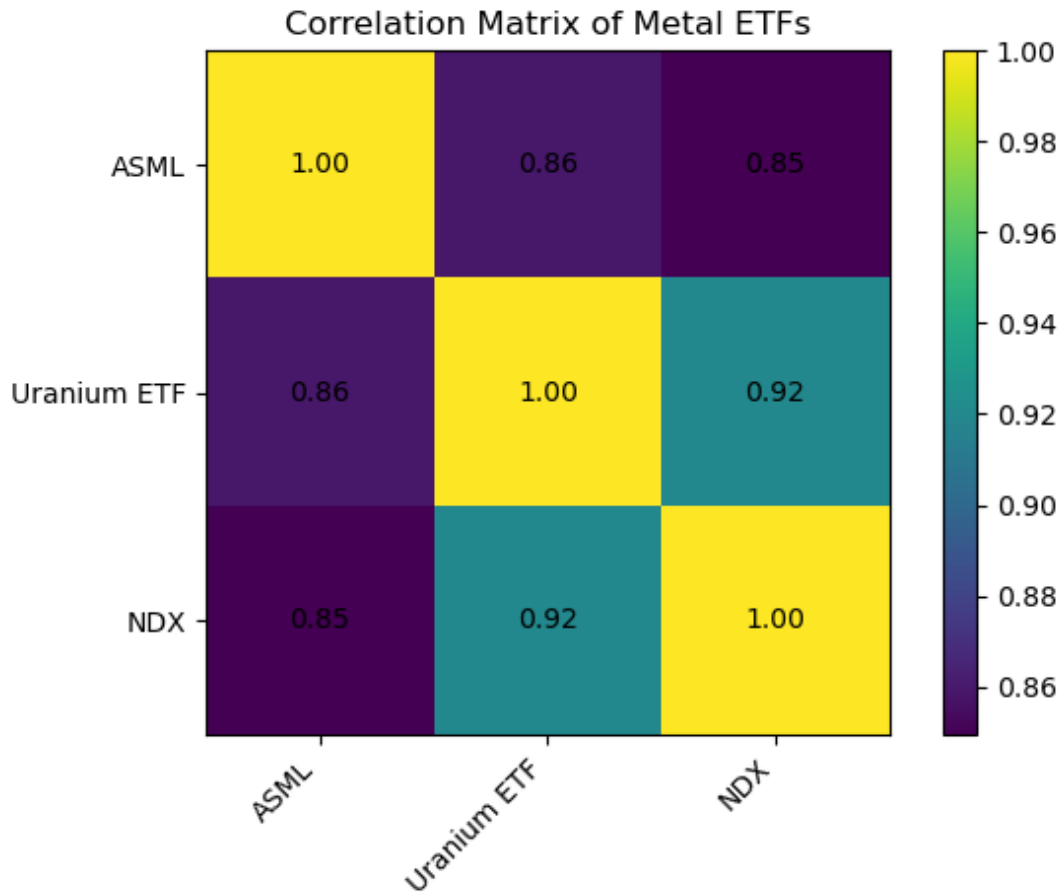


Figure 13: Correlation matrix of Model Composite parameters: ASML, Uranium ETF, and NASDAQ 100

The high VIFs for the uranium ETF and Nasdaq 100 could be due to the high correlation coefficient at 0.92, but it could also be attributed to the fact that most, if not all, of the companies in the nasdaq 100 (tech companies) depend on the energy sector to exist.

These parameters best explain NVIDIA's historical price movements ($R^2 = 0.932$), with uranium's dominance highlighting how NVIDIA's valuation has evolved into a broader bet on AI infrastructure rather than traditional semiconductor fundamentals.

4 Conclusion

4.1 Model Results

Overall, the models can be divided into three tiers. Tier 1, good models, includes only model 2; with R^2 of 0.815, the best Durbin-Watson result (2.586, $p = 0.29$), and good Breusch-Pagan and White’s test results—its only weakness being high VIF scores (See 3.1.2). The next tier includes models that, after the Cochrane-Orcutt procedure, returned decent values. Included in this tier are model 7 and model composite; these had severe autocorrelation but after running through the Cochrane-Orcutt procedure they returned much better results. Their Durbin-Watson statistics improved to 2.0387 and 2.0507 respectively. However, the adjusted R^2 s are much lower after correction, at 0.4613 and 0.4459 respectively; the models still have good Breusch-Pagan and White’s tests and acceptable VIFs (See 3.1.7 and 3.1.8). The next tier includes all models with a good R^2 but severe autocorrelation. Models 1, 3, 4, 5, and 6 all have good R^2 , ranging from 0.898 to 0.932, Breusch-Pagan and White’s test results, and adequate VIFs, but their performance is greatly hindered by their results of the Durbin-Watson test, ranging from 0.034 to 0.170. The Cochrane-Orcutt correction was used to correct for these low DW statistics, but this would result in near 0 R^2 . (See 3.1.1, 3.1.3, 3.1.4, 3.1.5, 3.1.6)

4.2 Answering the Research Question

Linear regression can reliably estimate NVIDIA’s stock close price only to a moderate extent. It was found that linear regression models are reliable only to a moderate extent, with the best corrected models (Model 7 and Model Composite) retaining adjusted R^2 values of 0.461 and 0.446 respectively after applying the Cochrane-Orcutt procedure — meaning linear regression can explain approximately 45% of NVIDIA’s price movements when autocorrelation is properly accounted for. This is likely because NVIDIA is a highly idiosyncratic stock, meaning NVIDIA’s price movements are driven by company-specific factors such as AI adoption and GPU demand rather than general market trends. This is consistent with Fama’s (1970) Efficient Market Hypothesis, which holds that

stock prices fully reflect all available information, making systematic estimation difficult. A more appropriate model for this type of data would be Autoregressive Integrated Moving Average (ARIMA), which is specifically designed for time series data. Unlike OLS, ARIMA models autocorrelation explicitly within its structure rather than requiring post-hoc corrections such as Cochrane-Orcutt or Newey-West, making it better suited for financial time series like NVIDIA's stock price.

5 Bibliography

Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2), 383–417.

Gujarati, D. N. (2011). *Econometrics by example*. Palgrave Macmillan.

Malhotra, N., & Tandon, K. (2013). *Determinants of stock prices: Empirical evidence from NSE 100 companies*. *International Journal of Research in Management & Technology*, 3(3), 86–90.

Oklo Inc. (n.d.). *About Oklo*. Oklo. <https://oklo.com/about/default.aspx>

R1B0PR07008. (2026). *Clean Data Fixed from Model 7.ipynb* (Version from School-Public repository) [Jupyter Notebook]. GitHub. <https://github.com/R1B0PR07008/School-Public/blob/main/Extended%20Essay/Clean%20Data%20Fixed%20from%20Model%207.ipynb>

Reuters. (2025, August 5). *Intel struggles with key manufacturing process for next PC chip, sources say*. Reuters. <https://www.reuters.com/world/asia-pacific/intel-struggles-with-key-manufacturing-process-next-pc-chip-sources-say-2025-08-05/>

U.S. Geological Survey. (n.d.). *Aluminum statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/aluminum-statistics-and-information>

U.S. Geological Survey. (n.d.). *Copper statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/copper-statistics-and-information>

U.S. Geological Survey. (n.d.). *Gold statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/gold-statistics-and-information>

U.S. Geological Survey. (n.d.). *Platinum-group metals statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/platinum-group-metals-statistics-and-information>

U.S. Geological Survey. (n.d.). *Rare earths statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/rare-earths-statistics-and-information>

ment of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/rare-earths-statistics-and-information>

U.S. Geological Survey. (n.d.). *Silicon statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/silicon-statistics-and-information>

U.S. Geological Survey. (n.d.). *Silver statistics and information*. U.S. Department of the Interior. <https://www.usgs.gov/centers/national-minerals-information-center/silver-statistics-and-information>

Wooldridge, J. M. (2012). *Introductory econometrics: A modern approach* (5th ed.). South-Western Cengage Learning.

Yahoo Finance. (n.d.). *Yahoo Finance*. Yahoo. <https://finance.yahoo.com/>